All Adder Neural Networks for On-Board Remote Sensing Scene Classification

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***Abstract*— Directly performing remote sensing scene classification (RSSC) on satellites can effectively relieve pressures on data downlinks. However, existing convolutional neural network (CNN)-based methods with dense multiplication operations require a vast resource overhead and can hardly be deployed on resource-limited space platforms. In this article, an all adder neural network (A^2NN) with a generative-based hybrid knowledge distillation (GHKD) training strategy is proposed to solve this problem. In contrast to traditional CNNs, the A^2NN discards all multiplication operations in the convolutional layers of the network and is instead completely composed of adder kernels; thus, this network has lower resource overheads than existing CNNs in hardware deployment. Compared with CNNs, the A^2NN has a considerably lower RSSC accuracy; fortunately, this can be mitigated with the proposed GHKD training strategy. This strategy combines a hybrid KD method and a knowledge-matching-based generative learning method. The hybrid KD method utilizes the weight distribution knowledge in the adder neural network (ANN) to guide the high-performance classification knowledge in the CNN to be smoothly transferred to the A^2NN, which consists entirely of adder kernels and has a considerably different structure than the CNN. The knowledge-matching-based generative learning method generates effective samples by matching the knowledge of the feature distribution of the real samples in the knowledge sources to augment the training database for hybrid KD. Extensive experiments and analyses on six public RSSC datasets show that the proposed GHKD training strategy significantly improves the RSSC performance of the proposed A^2NN, achieving comparable performance to CNNs on most datasets.**

***Index Terms*— Deep learning, generative learning, knowledge distillation (KD), on-board processing, scene classification.**

I. Introduction

REMOTE sensing scene classification (RSSC) is a basic remote sensing imagery interpretation task [1], [2], [3]. In recent years, convolutional neural networks (CNNs) have exhibited powerful feature abstraction abilities in scene classification [4], [5]. Therefore, CNN-based RSSC methods have gradually surpassed hand-designed feature methods and have become a focus of RSSC research [6], [7].

In traditional remote sensing, remote sensing images are first downloaded from satellites to ground stations and then interpreted and analyzed to perform RSSC [8], [9]. However, with the development of remote sensing technology, the resolution and amount of data in remote sensing images have rapidly escalated, increasing the pressure on data downlinks [10], [11]. Thus, the application of CNN models on satellite edge devices for onboard processing is an intuitive solution [12], [13]. However, existing CNN-based RSSC methods that demonstrate excellent classification performance (e.g., ResNet18 [14] and VGGNet-13 [15]) require billions of multiplication and addition operations, which are difficult to deploy directly on the edge devices of space platforms, such as field-programmable gate arrays (FPGAs) and application-specific integrated circuits (ASICs) [1], [16].

Many researchers focus on reducing the model complexities of CNNs. The most common methods are network structure compression [17], [18] and quantization [16], [19]. Extensive studies in the first category explored compact network architectures to construct efficient neural networks and decrease computation. Li et al. [17] proposed a network pruning method to remove redundant weights, thus compressing the original network. Han et al. [18] proposed discarding subtle weights in pretrained deep networks by learning only the important connections to omit their original calculations. However, the criteria for compression need to be subtly set manually in these network structure compression methods to avoid severe accuracy drops. The studies in the second category are dedicated to reducing the computation complexity by converting the floating-point numerical of the models into a low-bit-width fixed-point numerical representation. Gupta et al. [19] proved the accuracy drop with the 16-bit fixed-point model was negligible. In addition, the INT8 models are widely used due to their high accuracy and moderate model size [16]. However, many studies show that multiplication requires more resources and time overheads than addition [20], [21]. Convolutional layers with billions of multiplications account for most of the calculations in CNN models [22]. Although both category methods can reduce the model complexities, these optimized models still rely on volumes of multiplications in computation. It is difficult to apply such models directly to on-board deployment.

Recently, Chen et al. [23] proposed an adder neural network (ANN) based on adder kernels. The ANN is a new

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deep-learning basic network that converts most of the multiplication operations of the convolution layers into addition operations, significantly reducing the resource overhead required for deployment on edge devices [21], [23]. However, the ANN retains the standard convolution kernels in the first and last layers. Thus, when an ANN is deployed on edge devices, the processing engines for the standard convolution kernel and adder kernel must be developed separately, increasing the hardware overheads. To resolve this issue, all standard convolution kernels in the ANN can be replaced with adder kernels, resulting in a novel basic network, all ANN (A^2NN). For the same backbone model, the network structure of these basic networks is consistent, and the difference is that the kernels to build the network are different. The A^2NN fundamentally solves the problem of high resource overhead caused by dense multiplication operations when deploying complex neural networks. Nevertheless, the feature extraction ability of the adder kernel lags behind that of the standard convolution kernel [23], which significantly limits the classification ability of an A^2NN model without any multiplication operations.

To address this issue, an A^2NN with a generative-based hybrid knowledge distillation (GHKD) training strategy is proposed in this article. The proposed A^2NN replaces all standard convolution kernels with adder kernels, resulting in a lower resource overhead than CNN and ANN in the inference phase, and is more suitable for deployment on edge devices. The GHKD training strategy for the A^2NN combines a hybrid KD method and a knowledge matching-based generative learning method, based on KD [24] and generative learning, respectively. KD is a teacher-guiding-student strategy in which a weak student network learns and imitates the output of a powerful teacher network, leading to the optimization of the student network [25]. Generative learning is a common method for expanding and augmenting the number of training samples [26]. In the proposed GHKD training strategy, the hybrid KD method utilizes an ANN with both adder kernels and standard convolution kernels to guide the knowledge in a high-performance CNN to be smoothly transferred to a student A^2NN model that consists entirely of adder kernels. The knowledge-matching-based generative learning method generates effective samples by matching the knowledge about the feature distribution of the real samples in the knowledge sources to augment the training database for hybrid KD.

The main contributions of this article can be summarized as follows.

1) A novel deep-learning basic network, A^2NN, with a GHKD training strategy suitable for onboard RSSC is proposed. The A^2NN replaces all standard convolution kernels with adder kernels, reducing the hardware resource consumption. The proposed GHKD training strategy combines a knowledge matching-based generative learning method and a hybrid KD method to mitigate the substantial decrease in accuracy of the A^2NN model for RSSC tasks. In the hybrid KD method, a well-trained ANN with both adder kernels and standard convolution kernels is introduced, which uses the generated samples obtained by the knowledge matching-based generative learning method to guide the knowledge in the high-performing CNN to be smoothly transferred to an A^2NN model using real-training samples.
2) A knowledge matching-based generative learning method is proposed to augment the number of training samples. By matching the knowledge about the feature distribution of the real samples in the knowledge sources, a generator is trained to generate fake samples with a similar data distribution as the real samples. The generated and real samples are used as two different training databases for hybrid KD.
3) A hybrid KD method is proposed to obtain an A^2NN model with excellent classification performance by maximizing the transfer of high-performance knowledge from well-trained models to the A^2NN model. Specifically, a well-trained ANN with a structure similar to the A^2NN model is introduced as a teacher to transfer knowledge on suitable weight distribution for the adder kernels to the A^2NN model. On this basis, a well-trained CNN is applied as another teacher to transfer high-performance classification knowledge to the A^2NN model. Guided by the knowledge in the well-trained ANN, the knowledge in the high-performance CNN can be smoothly transferred to the A^2NN model, whose structure is considerably different from that of the CNN.

To evaluate the effectiveness of the proposed strategy, exhaustive experiments are conducted on six public RSSC datasets. The experimental results show that the proposed training strategy can be used to develop resource-efficient and high-performing A^2NN models for onboard RSSC.

The remainder of this article is organized as follows. Section II briefly introduces some related works. Section III discusses preliminary knowledge about the adder kernel and the ANN. Section IV presents the details of the proposed A^2NN with the GHKD training strategy. Section V provides detailed experimental results on six public datasets. Section VI concludes this article.

II. Related Works

In this section, we briefly review related works from three aspects: RSSC, generative learning, and KD.

A. Remote Sensing Scene Classification

RSSC focuses on refining semantic-level information and assigning a category label to the given image, which has attracted wide attention in the field of remote sensing imagery interpretation.

Various methods have been proposed for RSSC, including hand-crafted feature-based methods [27], [28], feature-encoding methods [29], [30], and deep learning-based methods [31], [32], [33], [34], [35], [36]. Hand-crafted feature-based methods, such as texture descriptors [28] and histograms of oriented gradient features, can only describe information in a single aspect or several aspects. Furthermore, the performances of these methods are limited by human experience [37]. In contrast, feature-encoding methods, such as bag-of-visual-words, probabilistic latent semantic analysis, and the

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latent Dirichlet allocation method, which can extract high-order statistical information for remote sensing scenes, are effective. However, these methods are still tricky to extract complex features because of their simple encoding architectures and the rich variations in remote sensing scenes.

Due to powerful feature abstraction and generalization abilities, deep learning-based methods, especially CNNs, have achieved tremendous development in RSSC. For example, Cheng et al. [31] presented a discriminative-CNN (D-CNN) trained by a new discriminative objective function that introduces metric learning to the current CNN structure. Because the feature distance between the same classes is small and that between different classes is large in the feature space of D-CNN, the performance of D-CNN was significantly improved. Wang et al. [32] presented a structured key area localization (SKAL) method to localize the most important area in remote sensing scene images. A global-local two-stream architecture was further presented based on SKAL to joint global and local feature representation, which somewhat solves the problem of large-scale variance in remote sensing scene images and achieves excellent classification performance. Li et al. [33] proposed a key filter bank network (KFBNet) that utilizes a key filter bank to capture discriminative local details and retain global information simultaneously and significantly improves the performance of CNN feature extractors. Wang et al. [34] presented a multigranularity canonical appearance pooling (MG-CAP) framework, which can automatically capture the latent ontologies in remote sensing scene datasets to improve performance. Recently, the transformer has shown outstanding computer vision performance and has been applied to the RSSC. Bazi et al. [35] introduced the vision transformer (ViT) model [38] into RSSC to boost the classification performance. Lv et al. [36] proposed a spatial-channel feature-preserving ViT (SCViT) model, which improves the discriminative ability of the ViT model by considering both the structural and channel information.

Although the above-mentioned advanced deep learning-based methods for RSSC can obtain outstanding classification performance, they are difficult to be deployed on edge devices due to their high model complexity and large parameters.

B. Generative Learning

Generative learning has made remarkable progress in recent years and can be broadly classified into several categories: autoregressive models, variational autoencoders (VAEs), and generative adversarial networks (GANs) [39], [40]. Among the existing generative learning methods, the GAN is the most widely applied method for generating samples [41], [42]. The generator and discriminator in the GAN iteratively optimize their parameters by competing and restricting one another, thus improving their abilities to generate and discriminate samples [43]. For RSSC, the GAN is a common method for generating remote sensing image scene samples to improve the accuracy of various classification models. For example, Lin et al. [44] first applied a GAN to RSSC. They proposed a multilayer feature matching GAN that learned interpretable representations from original datasets to improve the classification accuracy. Yu et al. [45] proposed an attention-GAN method that improved the classification performance by enhancing the representation ability of the discriminator. Ma et al. [46] proposed a progressive growing sample generation method for generating high-quality scene classification samples for RSSC. However, training a GAN model with an excellent generative ability for RSSC is complicated [31], [47].

Some GAN-based data-free generation methods have been proposed in fields with high data privacy or confidentiality requirements [48], [49]. These methods do not use original images or the adversarial process while training generative models; instead, they use well-trained models to guide the generator to generate new samples. Training the generator to achieve convergence is considerably easier with these methods than training native GANs to achieve convergence. Inspired by these GAN-based data-free generation methods, a low training difficulty generative learning method is proposed in this article to generate effective samples to augment training databases without using real samples or the adversarial process. This novel method is then used to alleviate the reduced accuracy of the A^2NN model.

C. Knowledge Distillation

The basic idea of KD is to distill knowledge from a pretrained teacher model to a student model by learning the class distributions provided by the teacher model. Recently, to develop lightweight networks with comparable performance to large networks, KD has been widely used to achieve model compression [50]. Hinton et al. [24] first proposed the KD method and found that the use of soft labels can improve the results of the student network more than the use of only hard labels. Following this finding, many works have focused on exploring how to encode the knowledge of teacher models and how to transfer this knowledge to student models, including the various forms of knowledge for distillation. Representative forms include attention maps [51], feature transformation [52], [53], sample relationships [54], [55], and activation boundaries [56].

In most existing KD methods, the lightweight student model learns knowledge from a pretrained teacher model with a similar architecture. Since the two models have similar information extraction methods, the knowledge of the teacher model can be smoothly transferred to the student model. Thus, the lightweight student model can inherit the excellent performance of the teacher model. However, since the A^2NN is constructed with novel adder kernels, these existing KD methods are problematic when they are used to directly distill knowledge from a teacher model to improve the A^2NN performance. Therefore, a hybrid KD method is proposed in this article to maximize the transfer of high-performance knowledge from the well-trained models to the A^2NN model. Specifically, a well-trained ANN with a similar structure is used as a teacher model to transfer knowledge on a suitable weight distribution for the adder kernels to the A^2NN model. On this basis, a well-trained CNN is introduced as another teacher to transfer high-performance classification knowledge to the A^2NN model. In other words, the weight distribution

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knowledge of the ANN can guide the high-performance classification knowledge in the CNN to be smoothly transferred to the A^{2}NN model, which has a completely different structure than the CNN.

III. Preliminary Knowledge

Since the proposed A^{2}NN consists of adder kernels, the adder kernel and the ANN are briefly reviewed and analyzed in this section.

A standard convolution can be defined as

$$O_{x,y,t} = \sum_{m=0}^{d}\sum_{n=0}^{d}\sum_{k=0}^{c_{\text{in}}} (I_{x+m,y+n,k} \times F_{m,n,k,t}) \tag{1}$$

where $I \in \mathbb{R}^{h_{\text{in}}\times w_{\text{in}}\times c_{\text{in}}}$ and $O \in \mathbb{R}^{h_{\text{out}}\times w_{\text{out}}\times c_{\text{out}}}$ represent the input feature tensor and the output feature tensor, respectively, and $F \in \mathbb{R}^{d\times d\times c_{\text{in}}\times c_{\text{out}}}$ represents the filter tensor. The convolution can be regarded as a similarity measure between the input features and filters. The output feature O can be formulated as follows:

$$O_{x,y,t} = \sum_{m=0}^{d}\sum_{n=0}^{d}\sum_{k=0}^{c_{\text{in}}} S(I_{x+m,y+n,k}, F_{m,n,k,t}) \tag{2}$$

where $S(\cdot)$ is a predefined similarity measure. To eliminate the multiplication operations, the adder kernel uses the $L1$ distance to indicate the similarity between the filters and the input features [23]. Thus, the adder kernel is formulated as follows:

$$O_{x,y,t} = -\sum_{m=0}^{d}\sum_{n=0}^{d}\sum_{k=0}^{c_{\text{in}}} |I_{x+m,y+n,k} - F_{m,n,k,t}|. \tag{3}$$

CNN models utilize backpropagation to compute the gradients of the filters and the stochastic gradient descent (SGD) to update the parameters. Since the adder kernel does not include multiplication, its partial derivative is a constant. Thus, the gradients of the adder kernel can be defined as follows:

$$\frac{\partial O_{x,y,t}}{\partial F_{m,n,k,t}} = I_{x+m,y+n,k} - F_{m,n,k,t} \tag{4}$$

$$\frac{\partial O_{x,y,t}}{\partial I_{x+m,y+n,k}} = \text{HTF}\left(F_{m,n,k,t} - I_{x+m,y+n,k}\right) \tag{5}$$

where HTF$(\cdot)$ is the HardTanh function, which is defined as follows:

$$\text{HTF}(x) = \begin{cases} -1, & x \le -1 \\ x, & -1 < x < 1 \\ 1, & x \ge 1. \end{cases} \tag{6}$$

Based on the adder kernels, Chen et al. [23] proposed an ANN that eliminates most convolution operations in the network. A comparison of the structures of the ANN and CNN models is shown in Fig. 1.

IV. Proposed Methods

In each training iteration of the GHKD training strategy, two well-trained networks for the RSSC, namely, a CNN and an ANN, are used as knowledge sources in the knowledge matching-based generative learning method to guide the optimization of the generator to generate effective fake samples for hybrid KD. Then, in the hybrid KD method, the above-mentioned well-trained CNN and ANN models are employed as teachers to guide the student A^{2}NN model by transferring the knowledge in their area of expertise to improve the classification performance of the A^{2}NN model.

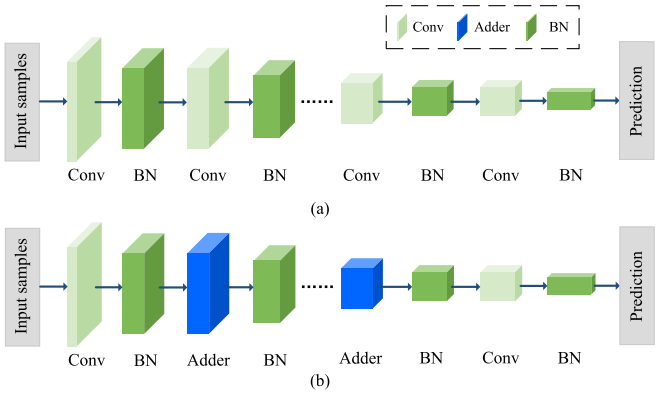

Fig. 1. Comparison of the structures of the ANN and CNN models. (a) Framework of the CNN. (b) Framework of the ANN.

A. Overview of the A^{2}NN With the GHKD Training Strategy

The proposed A^{2}NN has a low resource overhead because all standard convolution kernels are replaced by adder kernels; thus, this basic network is suitable for deployment on edge devices. The framework of the A^{2}NN is shown in Fig. 2(a). The A^{2}NN has a reduced accuracy for RSSC tasks, which can be mitigated using the proposed GHKD training strategy. This strategy combines a knowledge-matching-based generative learning method and a hybrid KD method. These two approaches are applied successively to optimize the generator and the A^{2}NN model, respectively, during one training iteration. The framework of the GHKD training strategy is presented in Fig. 2(b).

In the mth iteration of the proposed GHKD training strategy, first, generator G_θ, where θ represents the parameters of the generator, creates fake samples $\hat{S}$ based on random noise n and the random label $\hat{l}$, as follows:

$$\hat{S} = G_\theta(n|\hat{l}), n \sim N(0,1), \quad \hat{l} \sim \mu(0,1,\ldots,M-1) \tag{7}$$

where M represents the number of classes of real samples and $(n|\hat{l})$ is a labeled latent vector that is obtained by embedding the preset random label $\hat{l}$ and the corresponding random noise n. The generator G_θ uses the previously labeled latent vector as an input to generate fake samples for the specified class. As a result, the generator G_θ is encouraged to focus on each class, which ensures that the number of generated samples for each class is not unbalanced.

Following this, $\hat{S}$ is input into the well-trained CNN and ANN models, which are trained with real samples of remote sensing scenes to obtain their classification logits $\hat{F}_{\text{CNN}}$ and $\hat{F}_{\text{ANN}}$ as follows:

$$\hat{F}_{\text{CNN}}, \hat{\mu}_i, \hat{\sigma}_i = f_{\text{CNN}}(\hat{S}) \tag{8}$$

$$\hat{F}_{\text{ANN}} = f_{\text{ANN}}(\hat{S}) \tag{9}$$

where $f_{\text{CNN}}(\cdot)$ and $f_{\text{ANN}}(\cdot)$ are the function sets of the well-trained CNN and ANN models, respectively, and $\hat{\mu}_i$ and $\hat{\sigma}_i$

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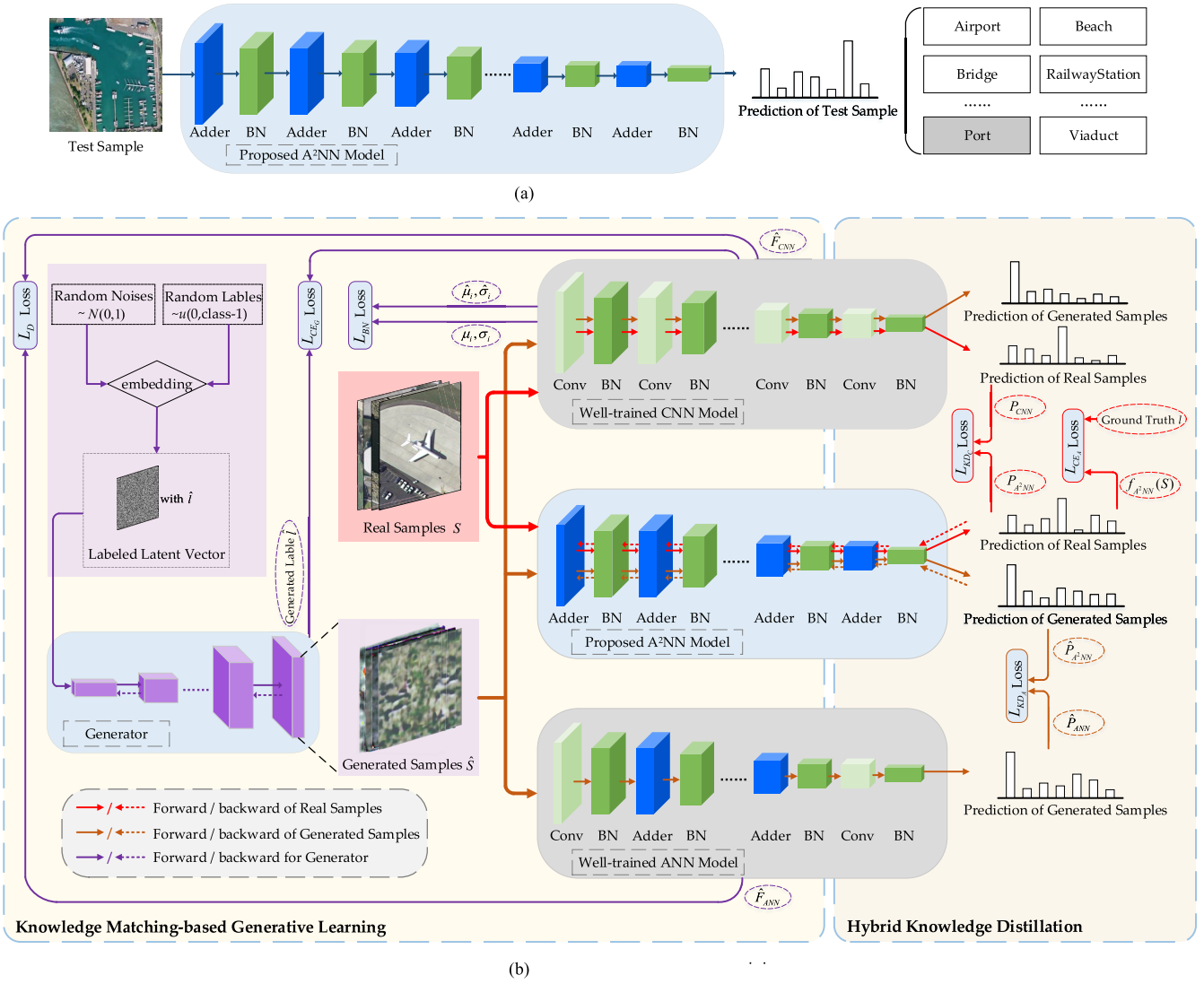

Fig. 2. Framework of the proposed A^{2}NN with the GHKD training strategy. (a) Network structure of the proposed A2NN. (b) Framework of the proposed GHKD training strategy.

are the mean and variance of the generated sample distribution in the ith batch normalization (BN) layer, respectively.

To expand the training database for the subsequent hybrid KD task, the generator G_θ matches the knowledge of the real samples maintained in the knowledge sources to generate new effective samples. To achieve this, the generator G_θ is optimized by constraining the intermediate results and classification logits shown in (8) and (9) in the loss function L_G as follows:

$$L_G = L_{\mathrm{CE}_G}(\hat{F}_{\mathrm{CNN}}) + \alpha L_{\mathrm{BN}}(\hat{\mu}_i, \hat{\sigma}_i) + \beta L_D(\hat{F}_{\mathrm{CNN}}, \hat{F}_{\mathrm{ANN}}) \tag{10}$$

where α and β are hyperparameters associated with balancing the three loss functions. The principle and details of the three loss functions used to construct the joint loss function L_G, namely, the cross-entropy loss of the generator L_{CE_G}, the BN loss L_{BN} and the distance loss L_D, are detailed in Section IV-B. The generator G_θ can be updated as follows:

$$\theta^{m+1} = \theta^m - lr_{G_m} \times \frac{\partial L_G}{\partial \theta^m} \tag{11}$$

where lr_{G_m} represents the learning rate of the generator G_θ in the mth iteration and θ^m and θ^{m+1} represent the parameters of the generator G_θ in the mth and $m+1$th iterations, respectively.

In the hybrid KD method, the well-trained ANN model and CNN model are introduced as teacher models, and the effective generated samples $\hat{S}$ and the real-training samples S are used as different training databases to transfer knowledge between the teacher models and the student A^{2}NN model. Due to its excellent classification performance, the teacher CNN model uses real-training samples S as the training database to guide the A^{2}NN model to improve classification ability. The classification logits of the teacher CNN can be calculated with the following equation:

$$F_{\mathrm{CNN}} = f_{\mathrm{CNN}}(S). \tag{12}$$

Since the adder kernels and the standard convolution kernels have different feature extraction methods and weight distributions, the potential of the student A^{2}NN model cannot be fully explored even though the teacher CNN model has excellent performance. Because both networks use adder kernels, ANN

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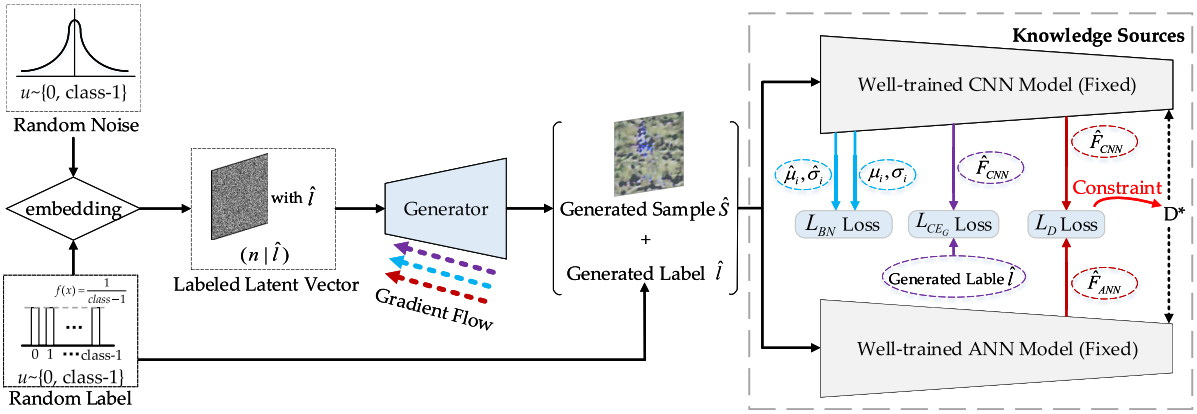

Fig. 3. Framework of the proposed knowledge matching-based generative learning method. The well-trained CNN and ANN models are used as knowledge sources to help the generator match the knowledge of the real samples and generate new samples with similar data distributions.

and A^{2}NN have similar feature extraction methods. Therefore, the teacher ANN model uses the generated samples $\hat{S}$ as its training database to guide the weight distribution of the A^{2}NN model. In this way, the ANN's weight distribution knowledge can guide the CNN's high-performance classification knowledge to be smoothly transferred to the student A^{2}NN model.

The loss function L_{HKD} is designed to optimize the A^{2}NN model using the ground truth of the real samples l and the prior knowledge of the two teachers: the high-performance classification knowledge of the teacher CNN F_{CNN} and the weight distribution knowledge of the teacher ANN $\hat{F}_{\mathrm{ANN}}$, as follows:

$$L_{\mathrm{HKD}} = L_{\mathrm{CE}_A}(l) + \lambda(L_{\mathrm{KD}_C}(F_{\mathrm{CNN}}) + L_{\mathrm{KD}_A}(\hat{F}_{\mathrm{ANN}})) \quad (13)$$

where λ is a hyperparameter for balancing the loss functions. The principle and details of the three loss functions used to construct the joint loss function L_{HKD}, namely, the cross-entropy loss of A^{2}NN L_{CE_A}, the KD loss of CNN L_{KD_C} and the KD loss of ANN L_{KD_A}, are detailed in Section IV-C. Thus, the A^{2}NN model can be updated as follows:

$$\gamma^{m+1} = \gamma^m - lr_{\mathrm{A^2NN}_m} \times \frac{\partial L_{\mathrm{HKD}}}{\partial \gamma^m} \quad (14)$$

where $lr_{\mathrm{A^2NN}_m}$ represents the learning rate of the A^{2}NN model in the mth training iteration, γ^m and γ^{m+1} represent the parameters of the A^{2}NN model in the mth and $m+1$th iterations, respectively.

The resulting A^{2}NN model trained by the GHKD training strategy has an excellent ability to extract discriminative features, which can be used to predict the final semantic class of the samples during the inference phase. Notably, the well-trained A^{2}NN model obtained by the GHKD training strategy does not add extra resource overhead in the inference phase.

B. Knowledge Matching-Based Generative Learning

A knowledge matching-based generative learning method is proposed to generate effective new samples with a similar data distribution as the real-training samples used as the training databases for the teacher ANN model in the hybrid KD. The details of the knowledge matching-based generative learning method are shown in Fig. 3, and its joint loss function L_G is shown in (10). L_{CE_G}, L_{BN}, and L_D are utilized to design the joint loss function L_G to optimize the generator G_θ.

The loss function L_{CE_G} is designed to minimize the distance between the classification logits of the knowledge sources and the preset classes $\hat{l}$ using the cross-entropy loss because effectively generated samples should be classified into preset classes according to well-trained knowledge sources. Therefore, the loss function L_{CE_G} can be expressed as follows:

$$L_{\mathrm{CE}_G} = \mathrm{CE}[\hat{F}_{\mathrm{CNN}}, \hat{l}]. \quad (15)$$

The loss function L_{BN} is designed to optimize the generator to ensure that the mean and variance of each BN layer obtained by the generated samples through the knowledge model match those of the real samples as much as possible. The BN layer [57] is used to stabilize the data distribution and learn the classification boundary, allowing the data to be divided into different classes. Thus, the BN layers can be used to obtain knowledge about the classification boundary and data distribution to guide the distribution of the generated samples. Therefore, the mean and variance of the data distribution can be introduced as quantitative descriptors of this knowledge. The loss function L_{BN} is defined as follows:

$$L_{\mathrm{BN}} = \frac{1}{N}\sum_{i=0}^{N}(\mathrm{MSE}(\hat{\mu}_i, \mu_i) + \mathrm{MSE}(\hat{\sigma}_i, \sigma_i)) \quad (16)$$

where N represents the number of BN layers in the CNN knowledge source, $\hat{\mu}_i$ and $\hat{\sigma}_i$ are the mean and variance of the generated sample distribution in the ith BN layer, respectively, μ_i and σ_i are the mean and variance of the real-sample distribution for the knowledge source stored in the ith BN layer, respectively, and $MSE(\cdot)$ is the mean squared error operator, which can be expressed as follows:

$$\mathrm{MSE}(\hat{\mu}_i, \mu_i) = \frac{1}{N}\sum_{i=0}^{N} \| \hat{\mu}_i - \mu_i \|_2^2. \quad (17)$$

The loss function L_D is designed to measure the difference between the classification boundaries of the well-trained CNN and ANN models obtained when the generated samples are used as input. Since the feature extraction ability of the

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CNN is better than that of the ANN, the above-mentioned two loss functions L_{CE_G} and L_{BN} are introduced to extract the classification boundary and distribution knowledge from the well-trained CNN model to optimize the generator G_θ. Furthermore, because the ANN and A^{2}NN models have similar network structures, some knowledge about the real samples can be extracted from the well-trained ANN model to ensure that the generated samples conform to the prior knowledge of the ANN knowledge source. By minimizing L_D, the generated samples are classified with the same label $\hat{l}$ by the CNN and ANN models. The loss function L_D is formulated as follows:

$$L_D = \log[\mathrm{MAE}(\hat{F}_{\mathrm{CNN}}, \hat{F}_{\mathrm{ANN}}) + 1] \tag{18}$$

where the log function is used to stabilize the training of the generator, and *MAE(·)* is the mean absolute error operator, which can be expressed as follows:

$$\mathrm{MAE}(\hat{F}_{\mathrm{CNN}}, \hat{F}_{\mathrm{ANN}}) = \frac{1}{N}\sum_{j=0}^{B} |f_{\mathrm{CNN}}(\hat{S})_j - f_{\mathrm{ANN}}(\hat{S})_j| \tag{19}$$

where B represents the batch size of the training.

By minimizing L_D, the feature distribution of the generated samples on both the CNN and ANN knowledge sources can gradually approach that of the real samples. In this way, the generated samples can be used as a valid training database for the teacher ANN model in hybrid KD.

C. Hybrid Knowledge Distillation

A hybrid KD method is proposed to improve the classification performance of the target network A^{2}NN. A well-trained CNN and ANN models are employed as teacher models to play their respective advantages. More specifically, the proposed hybrid KD method uses the weight distribution knowledge of the ANN to guide the high-performance classification knowledge in the CNN to be smoothly transferred to the A^{2}NN model, which has a completely different structure than the CNN. The joint loss function L_{HKD} of the hybrid KD method is shown in (13). L_{CE_A}, L_{KD_C}, and L_{KD_A} are utilized to design the joint loss function L_{HKD} to optimize the A^{2}NN model.

The loss function L_{CE_A} is designed to minimize the distance between the classification logits of the A^{2}NN model and the ground truth. An A^{2}NN model with excellent classification performance should demonstrate high classification accuracy on real-training samples. The cross-entropy loss is used to build the loss function L_{CE_A}, which can be expressed as follows:

$$L_{\mathrm{CE}_A} = CE[f_{\mathrm{A^2NN}}(S), l] \tag{20}$$

where $f_{\mathrm{A^2NN}}(\cdot)$ is the function set of the A^{2}NN model and l represents the ground truth of the real samples.

The loss function L_{KD_C} is designed to optimize the A^{2}NN model to ensure that the student A^{2}NN model imitates the classification boundary of the teacher CNN model on the real-training samples. Because the A^{2}NN model has a weak feature extraction ability, the knowledge this model can extract through the hard labels of real samples is limited. Therefore, real samples are used as the training database, and a well-trained CNN model with excellent classification accuracy is applied as a teacher to guide the student A^{2}NN through the soft labels to further improve performance. The Kullback–Leibler (KL) divergence is introduced to enable the A^{2}NN model to imitate the classification boundary of the teacher CNN model. The L_{KD_C} is formulated as follows:

$$\begin{aligned} L_{\mathrm{KD}_C} &= T^2 \cdot \mathrm{KLDiv}[P_{\mathrm{A^2NN}}, P_{\mathrm{CNN}}] \\ &= T^2 \cdot \sum_{k=0}^{M} P_{\mathrm{A^2NN}}(k) \log \frac{P_{\mathrm{A^2NN}}(k)}{P_{\mathrm{CNN}}(k)} \end{aligned} \tag{21}$$

where T is a hyperparameter that determines the soft degree of labels, with a higher value of T resulting in a softer probability distribution over the classes; $P_{\mathrm{A^2NN}}$ is the class probability generated by using a log_softmax function with T for the A^{2}NN model; and P_{CNN} is the class probability generated by using a softmax function with T for the CNN model. The kth elements of $P_{\mathrm{A^2NN}}$ and P_{CNN} are computed as follows:

$$P_{\mathrm{A^2NN}}(k) = \log \frac{e^{\frac{\mathrm{A^2NN}(k)}{T}}}{\sum_{b=0}^{M} e^{\frac{\mathrm{A^2NN}(b)}{T}}} \tag{22}$$

$$P_{\mathrm{CNN}}(k) = \frac{e^{\frac{CNN(k)}{T}}}{\sum_{b=0}^{M} e^{\frac{CNN(b)}{T}}}. \tag{23}$$

Notably, the feature abstraction methods of the standard convolution kernels and adder kernels are different. The standard convolution kernels use the ℓ_2-distance to measure differences between the input features and filters; the adder kernels use the ℓ_1-distance. Moreover, the weights in well-trained standard convolution kernels are often Gaussian distributions, while the weights in well-trained adder kernels are often Laplacian distributions [23]. Because CNNs and A^{2}NNs have different feature extraction methods and weight distributions, the potential of the student A^{2}NN model cannot be fully explored even though the teacher CNN model has excellent performance. Thus, a well-trained ANN with a similar network structure to the A^{2}NN is employed as another teacher to guide the student A^{2}NN model with the knowledge of weight distribution.

The loss function L_{KD_A} is designed to optimize the A^{2}NN model to ensure that the student A^{2}NN model imitates the weight distribution of the teacher ANN model. Similarly, the KL distance is used to optimize the student model. The L_{KD_A} is denoted as follows:

$$\begin{aligned} L_{\mathrm{KD}_A} &= T^2 \cdot \mathrm{KLDiv}[\hat{P}_{\mathrm{A^2NN}}, \hat{P}_{\mathrm{ANN}}] \\ &= T^2 \cdot \sum_{k=0}^{M} \hat{P}_{\mathrm{A^2NN}}(k) \log \frac{\hat{P}_{\mathrm{A^2NN}}(k)}{\hat{P}_{\mathrm{ANN}}(k)} \end{aligned} \tag{24}$$

where $\hat{P}_{\mathrm{A^2NN}}$ and $\hat{P}_{\mathrm{ANN}}$ are the class probabilities obtained from the A^{2}NN and ANN models using generated samples as input. In particular, since the ANN has a certain gap with CNN in the classification performance, the use of the same real samples S as the training database for the teacher ANN model has an adverse effect on the knowledge transfer effect of the teacher CNN model. Therefore, the generated samples $\hat{S}$ obtained earlier during the current iteration are used as the training database to transfer the knowledge of the teacher ANN model.

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TABLE I
DETAILED INFORMATION OF SIX POPULAR DATASETS FOR GHKD

Datasets	UCM	WHU	RSSCN7	RS_C11	SIRI-WHU	AID
Classes	21	19	7	11	12	30
Number of images per class	100	∼50	400	∼100	200	220∼420
Total number of images	2100	1005	2800	1232	2400	10000
Spatial resolution (m)	0.3	up to 0.5	-	0.2	2	0.5∼8
Image size	256×256	600×600	400×400	512×512	200×200	600×600
Data source	USGS	Google Earth	Google Earth	Google Earth	Google Earth	Google Earth
Published year	2010	2012	2015	2016	2016	2017

Thus, the proposed hybrid KD method employs two well-trained models to help improve the performance of the student A^2NN model by exerting the strengths of both well-trained models. More specifically, the weight distribution knowledge of the ANN can guide the high-performance classification knowledge in the CNN to be smoothly transferred to the A^2NN model, which has a completely different structure than the CNN.

V. EXPERIMENTAL EVALUATION AND RESULTS

To evaluate the effectiveness of the proposed GHKD training strategy for the A^2NN, six public remote sensing datasets are employed for exhaustive experiments. The experiments are implemented using PyTorch 1.8 [58] and an NVIDIA Titan RTX graphical processing unit (GPU) for acceleration on a workstation with an Intel Xeon Gold 5120 CPU. Our experiments are described in detail in this section.

A. Datasets Description

In this article, UC-Merced Land Use Dataset (UCM) [59], WHU-RS19 (WHU) [60], RSSCN7 [61], RS_C11 [62], SIRI-WHU [30], and Aerial Image Dataset (AID) [63] are selected to conduct the following experiments. These six widely used datasets contain dozens of scene classes and have high within-class diversity, interclass similarity, and rich image variations, making the task of correctly recognizing these image scenes challenging. The key information of these six datasets is summarized in Table I.

Following recent research [64], [65], [66], for each dataset, a certain proportion of samples is randomly selected as the training database, and the remaining samples are used as the testing set. For the UCM, WHU, and RS_C11 datasets, 80% of the samples are randomly sampled for training and 20% of the samples are used for testing. For the SIRI-WHU dataset, 40% of the samples are randomly sampled for training, and 60% of the samples are used for testing. For the RSSCN7 and AID datasets, 20% of the samples are used for training, and the remaining 80% of samples are used for testing.

B. Evaluation Metrics

Following recent works on RSSC [64], [65], two generally accepted methods are adopted to comprehensively evaluate the classification performance of the proposed method: the overall accuracy (OA) and the confusion matrix.

The OA reflects the overall classification performance of the method and can be represented as follows:

$$\mathrm{OA} = \frac{L_c}{L_T} \tag{25}$$

where L_c is the number of test images that are correctly classified and L_T is the total number of test images.

A confusion matrix is a fine-grained table that reflects the detailed confusion degree and between-class classification errors achieved by a model. In a confusion matrix, the number of samples in each scene class that are correctly classified and misclassified as belonging to other scene classes can be intuitively observed. The entry in the pth row and qth column of the confusion matrix is calculated as follows:

$$C_{p,q} = \frac{T_{p,q}}{T_p} \tag{26}$$

where $T_{p,q}$ denotes the number of test images that belong to the pth class that are classified as the qth class, and T_p represents the total number of test images in the pth class. If $p = q$, the test samples are correctly classified; otherwise, the test samples are incorrectly classified.

C. Experimental Settings

1) Data Preprocessing: Following recent research [67], [68], [69], we apply data augmentation methods to the real samples in the experiments. We first resize all images of each dataset to 256 × 256 pixels. During training, the samples are randomly cropped to 224 × 224 pixel patches and flipped horizontally. During testing, the samples are center cropped to 224 × 224 patches.

2) Network: The experiments utilize three representative neural networks as classification network backbones: ResNet18, VGGNet-11, and VGGNet-13. To prevent redundancy, only the last fully connected (FC) layer in VGGNet-11 and VGGNet-13 is utilized as the classifier. According to (3), the output of the adder kernels is always negative. Hence, BN is introduced to normalize the output of the adder layers to an appropriate range [23]. To ensure the consistency of the structure, the BN layers are added after all the convolution layers in the three backbones. Moreover, the FC layers in the three backbones are replaced with 1 × 1 convolution layers to facilitate deployment.

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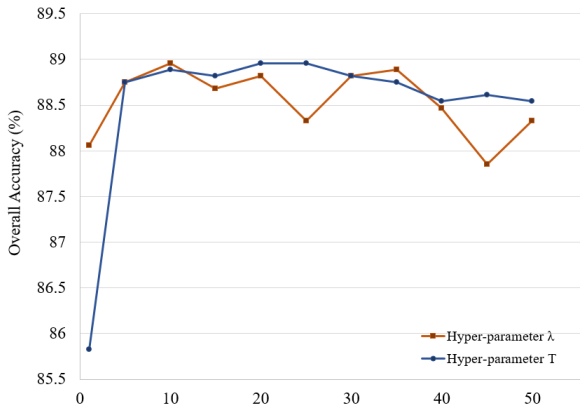

Fig. 4. OA for different values of the two hyperparameters on the SIRI-WHU dataset for the A^{2}NN-ResNet18 model.

On this basis, five models are developed for each backbone for our experiments. Based on the ResNet18 backbone, the CNN-ResNet18 model, binary neural network (BNN)-ResNet18 model, ANN-ResNet18 model, A^{2}NN-ResNet18 model, and A^{2}NN-ResNet18 model trained by the proposed GHKD strategy, denoted as the A^{2}NN-ResNet18_GHKD model, are developed. Based on the VGGNet-11 backbone, the CNN-VGGNet-11 model, BNN-VGGNet-11 model, ANN-VGGNet-11 model, A^{2}NN-VGGNet-11 model, and A^{2}NN-VGGNet-11 model trained by the proposed GHKD strategy, denoted as the A^{2}NN-VGGNet-11_GHKD model, are developed. Based on the VGGNet-13 backbone, the CNN-VGGNet-13 model, BNN-VGGNet-13 model, ANN-VGGNet-13 model, A^{2}NN-VGGNet-13 model, and A^{2}NN-VGGNet-13 model trained by the proposed GHKD strategy, denoted as the A^{2}NN-VGGNet-13_GHKD model, are developed. The BNN [70] is a lightweight basic network that uses only two possible values to constrain the weights, drastically reducing the number of multiplication operations. Furthermore, none of the models used in our experiments are loaded with pretrained models to ensure fairness.

3) Hyperparameter Settings: The five versions of the models described earlier used the same training settings on each dataset and are trained by SGD with a momentum of 0.9, a weight decay of 0.0005, and a batch size of 32. These models are trained for 300 epochs using the cosine learning rate decay [71] with an initial learning rate of 0.05. The generator is trained using Adam [72] with a learning rate of 0.001. Furthermore, the generator is trained for 50 epochs as a warm-up process before introducing the teacher ANN of the GHKD training strategy.

D. Hyperparameter Analysis

For the proposed hybrid KD method, two hyperparameters related to the loss function are crucial for determining the classification performance: the balance coefficient λ shown in (13) and the temperature T shown in (21) and (24). In this section, parameter sensitivity experiments are conducted on the SIRI-WHU dataset and the ResNet18 model to analyze their influence on the proposed training strategy and select the optimal values of the hyperparameters. The selected appropriate values of the hyperparameters of T and λ are also applied to the other five datasets and the other network in the following experiments.

To explore the performance of the proposed method with different values of the hyperparameter T, T is tuned according to the given set $\{1, 5, 15, \ldots, 50\}$ and the hyperparameter λ is fixed to 10. The variation in the OA with T is illustrated in Fig. 4. According to Fig. 4, the OA first increases dramatically and then stabilizes. This is because, for small values of T, the student model focuses on hard labels and fails to distill the a priori knowledge of the teacher models [64]. Furthermore, the proposed method obtains a satisfactory OA for a wide range of T, and the best classification result is achieved when T reaches 20 in the given set. Therefore, the value of T is set to 20 in the following experiments.

Similarly, λ is tuned to $\{1, 5, 15, \ldots, 50\}$ and the hyperparameter T is fixed to 20 to validate the OA for different values of the hyperparameter λ. The variation in the OA with the hyperparameter λ is illustrated in Fig. 4. The OA first gradually increases with increasing λ, reaching a peak value. When λ exceeds a certain value, the OA decreases slightly. The reason for this result is that an appropriate value of λ can balance the importance of the hard labels of the samples with the soft knowledge of the teacher model. Thus, the λ is set to 10 in the following experiments.

Furthermore, two hyperparameters, α and β, which are related to the loss function L_G, are crucial for ensuring the quality of the generated samples. Similarly, the grid search method is used to analyze the influence of these hyperparameters on the proposed training strategy and to select their optimal values. It is worth mentioning that the optimal values of these two hyperparameters are different for the various datasets due to differences in the image resolution and the information extracted by the well-trained models. The settings of α and β for the six datasets are shown in Table II.

TABLE II
SETTINGS OF THE HYPERPARAMETERS FOR DIFFERENT DATASETS

Datasets	UCM	WHU	RSSCN7	RS_C11	SIRI-WHU	AID
α	0.1	0.002	0.1	0.1	0.1	1
β	0.1	2	0.5	0.5	0.5	0.5

E. Comparison With Other Approaches

To illustrate the advancement and effectiveness of the proposed GHKD training strategy for A^{2}NN models, exhaustive experiments on six public RSSC datasets are conducted using ResNet18, VGGNet-11, and VGGNet-13 as the classification network backbones. Table III shows the classification results on the six datasets when different basic networks are used. All the listed models employ the same datasets with the same training ratios to ensure a fair comparison. Furthermore, all experiments are repeated five times to verify the reliability of the classification results. In addition, to compare the computational resource overheads of different basic networks during the inference phase, the number of operations used in the above fifteen models are counted and shown

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TABLE III

CLASSIFICATION ACCURACIES OF THE A^{2}NN MODEL TRAINED BY THE PROPOSED GHKD TRAINING STRATEGY AND OTHER BASIC NETWORKS WITH THE SAME ARCHITECTURES BY USING RESNET18, VGGNET-11, AND VGGNET-13 ON THE UCM, WHU, RSSCN7, RS_C11, SIRI-WHU, AND AID DATASETS [OA $\pm$ STD (%)]

Backbone	Basic Network	Datasets					
		UCM	WHU	RSSCN7	RS_C11	SIRI-WHU	AID
ResNet18	CNN	96.00±0.66	92.58±0.87	84.16±0.60	94.94±0.36	91.90±0.24	85.29±0.56
	BNN [70]	50.67±0.92	60.29±1.12	62.70±0.49	69.24±1.29	57.07±1.45	42.67±0.22
	ANN [23]	94.14±0.93	87.22±0.63	79.98±0.98	91.97±0.85	86.22±0.91	77.58±0.63
	A^{2}NN (Ours)	92.62±0.58	79.12±0.63	78.63±0.61	91.73±0.54	85.59±0.92	75.48±0.82
	A^{2}NN_GHKD (Ours)	95.62±0.27	90.24±0.49	82.42±0.60	93.49±0.34	89.61±0.24	79.15±0.32
VGGNet-11	CNN	96.10±0.89	91.90±0.74	82.12±0.42	94.46±0.60	87.53±0.40	83.28±0.59
	BNN [70]	89.09±0.57	82.73±1.56	78.03±0.62	90.12±1.12	80.56±0.21	66.65±0.85
	ANN [23]	94.67±0.46	89.46±0.74	79.98±0.82	93.41±0.46	86.68±0.73	81.28±0.65
	A^{2}NN (Ours)	94.05±0.29	87.22±1.31	78.92±0.93	92.05±0.92	85.90±0.67	79.56±0.83
	A^{2}NN_GHKD (Ours)	96.76±0.55	92.30±0.80	83.29±0.45	95.42±0.54	88.80±0.61	83.46±0.23
VGGNet-13	CNN	95.57±0.36	90.83±0.54	81.94±1.27	94.38±0.49	87.99±0.63	84.28±0.88
	BNN [70]	88.09±1.43	80.98±2.78	76.79±1.23	88.59±1.94	78.40±0.96	66.37±1.49
	ANN [23]	94.14±0.99	87.22±1.56	77.16±1.26	91.00±1.29	86.06±1.57	80.25±0.71
	A^{2}NN (Ours)	93.95±0.62	86.83±0.35	76.86±1.02	90.52±0.83	85.57±0.79	79.76±0.91
	A^{2}NN_GHKD (Ours)	96.53±0.21	90.93±0.44	81.70±0.71	92.77±0.57	88.88±0.18	84.20±0.39

TABLE IV

NUMBER OF OPERATIONS USED IN DIFFERENT BASIC NETWORKS

Backbone	Basic Network	FLOPs	Convolution/Adder/Binarize Layers			Other Layers
			# Mul.	# Add.	XNOR	MACs
ResNet18	CNN	3.63G	1.81G	1.81G	0	4.98M
	BNN	3.63G	0	1.81G	1.81G	4.98M
	ANN	3.63G	59.01M	3.56G	0	4.98M
	A^{2}NN (Ours)	3.63G	0	3.62G	0	4.98M
	A^{2}NN_GHKD (Ours)	3.63G	0	3.62G	0	4.98M
VGGNet-11	CNN	15.00G	7.49G	7.49G	0	14.85M
	BNN	15.00G	0	7.49G	7.49G	14.85M
	ANN	15.00G	43.36M	14.93G	0	14.85M
	A^{2}NN (Ours)	15.00G	0	14.97G	0	14.85M
	A^{2}NN_GHKD (Ours)	15.00G	0	14.97G	0	14.85M
VGGNet-13	CNN	22.42G	11.18G	11.18G	0	24.48M
	BNN	22.42G	0	11.18G	11.18G	24.48M
	ANN	22.42G	43.36M	22.33G	0	24.48M
	A^{2}NN (Ours)	22.42G	0	22.37G	0	24.48M
	A^{2}NN_GHKD (Ours)	22.42G	0	22.37G	0	24.48M

in Table IV. The other layers mainly refer to BN layers and activation layers. The computational costs of these layers are significantly less than those of the convolution/adder/binarize layers and can, thus, be ignored. The computational resource overhead of these models is mainly concentrated in the convolution/adder/binarize layers.

As shown in Table III, the A^{2}NN_GHKD models achieve nearly the same classification accuracies as those of traditional CNN models on most RSSC datasets and even surpass the classification accuracies of the CNN models on some datasets. For example, the A^{2}NN-VGGNet-11_GHKD models improve the OA by nearly 0.96%, 1.17%, and 1.27% on the RS_C11, RSSCN7, and SIRI-WHU datasets, respectively. Moreover, according to the resource overheads listed in Table IV, the A^{2}NN_GHKD models with no multiplication operations are, thus, suitable for deployment on edge devices with low resource costs.

According to Table III, the classification accuracy of the A^{2}NN_GHKD models exceeds that of the ANN models for all the listed datasets. Compared with the ANN models, for the ResNet18 backbone, the A^{2}NN-ResNet18_GHKD models improve the OA by nearly 1.48%–3.39%; for the VGGNet-11 backbone, the A^{2}NN_GHKD models improve the OA by nearly 2.01%–3.31%; for the VGGNet-13 backbone, the A^{2}NN_GHKD models improve the OA by nearly 1.77%–4.54%. The reason for this result is that in addition to learning from the teacher ANN models, the student model training with the proposed GHKD strategy benefits from the teacher CNN models. This learning experience enables the student A^{2}NN_GHKD models to surpass the ANN models with excellent feature extraction capabilities. Furthermore, the A^{2}NN_GHKD model replaces all standard convolution kernels of the ANN model with adder kernels, and the remaining multiplication operations of the ANN model are converted to addition operations in the A^{2}NN_GHKD model, as shown in Table IV.

According to Tables III and IV, although the BNN models are considerably smaller than the A^{2}NN and CNN models, their accuracy is substantially lower. For example, the BNN_ResNet18 model achieved an accuracy of only 50.67% on the UCM dataset. In contrast, the A^{2}NN-ResNet18_GHKD model achieved an accuracy of 95.62%. The reduced accuracy of the BNN models is unacceptable for RSSC tasks. Thus, the BNN models are not applicable for onboard RSSC.

In addition, the computing resource overhead required by the A^{2}NN and A^{2}NN_GHKD models during the inference phase is identical. Therefore, the resource-efficient and

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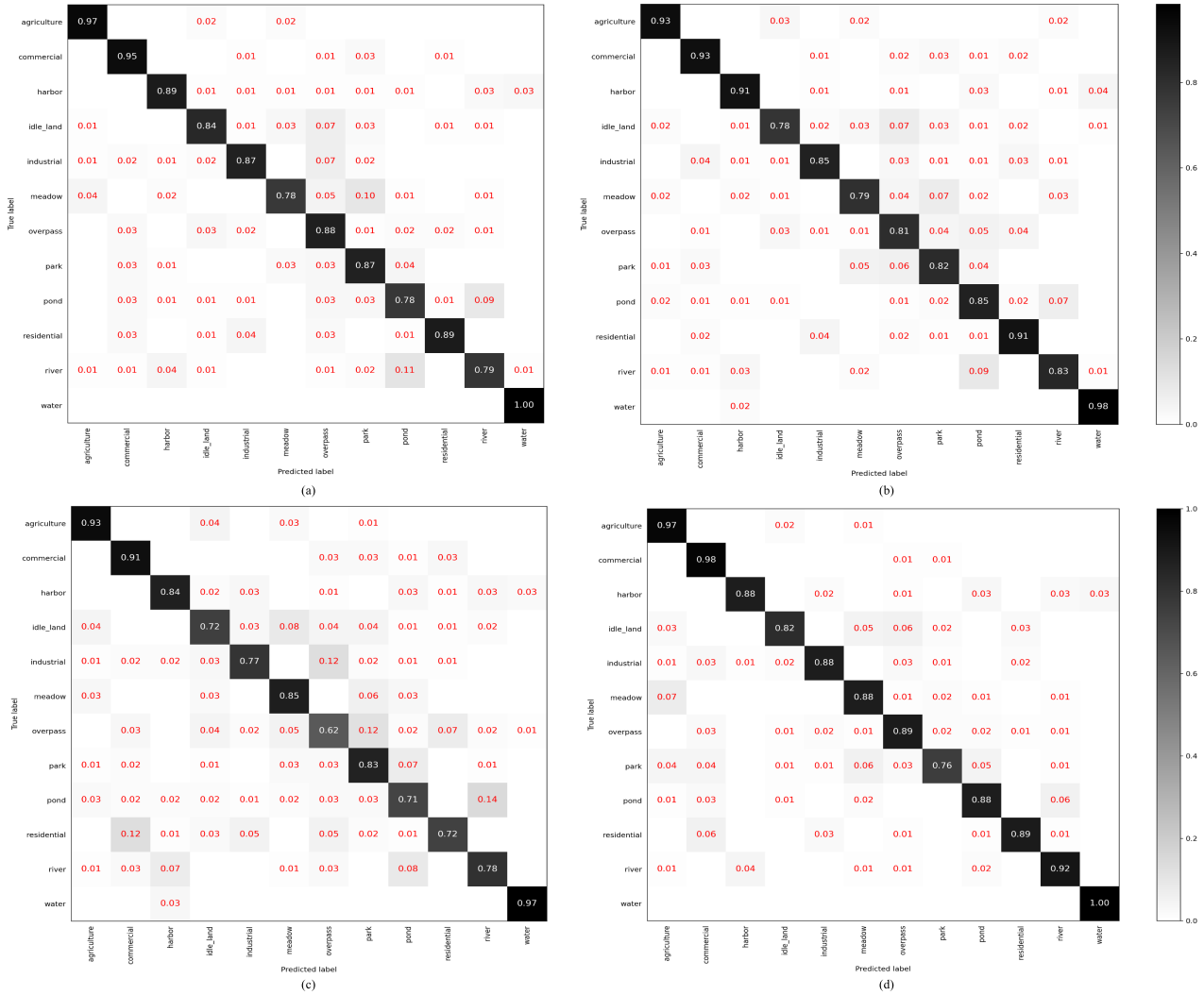

Fig. 5. Confusion matrices of different models based on the VGGNet-11 network on the SIRI-WHU dataset. (a) CNN-VGGNet-11 model. (b) ANN-VGGNet-11 model. (c) BNN-VGGNet-11 model. (d) A^{2}NN-VGGNet-11_GHKD model (ours).

high-performing A^{2}NN_GHKD models are better suited for deployment during the inference phase of RSSC.

The confusion matrices on the SIRI-WHU dataset using the VGGNet-11 network are reported in Fig. 5. Confusion matrices are provided for four different models: the CNN model, the ANN model, the BNN model, and the A^{2}NN_GHKD model. As shown in Fig. 5, the A^{2}NN-VGGNet-11_GHKD model shows an impressive performance. Specifically, the accuracies of ten classes are greater than 85%, and the total number of classes in the SIRI-WHU dataset is 12. In contrast, the ANN-VGGNet-11 and BNN-VGGNet-11 models, exhibit accuracies greater than 85% for only five and three classes, respectively. Furthermore, the CNN-VGGNet-11 model exhibits an accuracy of greater than 85% for only eight classes, which is also less than that of the A^{2}NN-VGGNet-11_GHKD model. The A^{2}NN-VGGNet-11_GHKD model achieves the best classification performance on this dataset among the four models, outperforming the CNN-VGGNet-11 model, the ANN-VGGNet-11 model, and the BNN-VGGNet-11 model by 1.27%, 2.12%, and 8.24%, respectively.

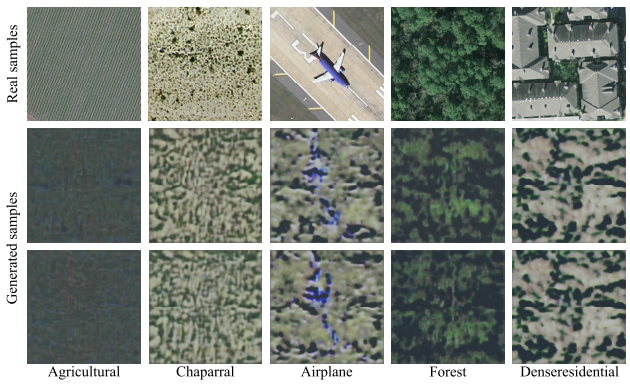

Fig. 6. Several classes of real samples and generated samples obtained by the GHKD method for the UCM dataset.

F. Generated Samples Visualization Analysis

In the proposed GHKD training strategy for the A^{2}NN models, the generated samples can enrich and augment the training database. Fig. 6 shows several classes of real samples and generated samples obtained by the GHKD method on the UCM dataset.

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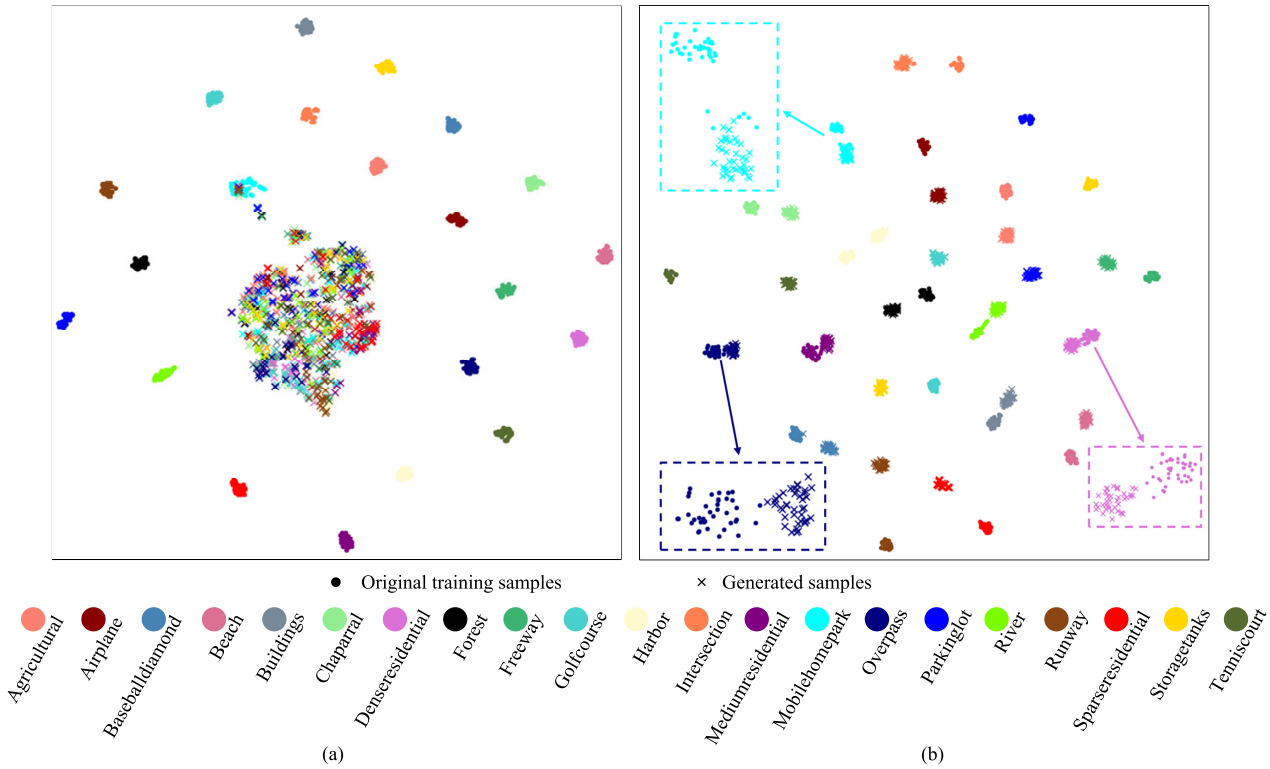

Fig. 7. Two-dimensional embedding of original training samples and generated samples on the UCM dataset with the CNN-ResNet18 model. (a) Epoch 1. (b) Epoch 250.

As shown in Fig. 6, for the remote sensing scene class of natural landmarks, such as agricultural land and forest, the generated samples are close to the real-scene samples. For the remote sensing scene class of artificial landmarks, such as airplanes and dense residential areas, the generated samples are far from being humanly interpretable, which mainly appears to be matching the classification knowledge about texture and color.

To explain and verify that the generated samples obtained by the knowledge matching-based generative learning method can be used to augment the training database, the t-SNE method [73] is employed to visualize 2-D embeddings to evaluate the distributions of the generated samples and original training samples in the embedding space. In this experiment, the generator is trained for 300 epochs on the UCM dataset using the ResNet18 backbone. At the first and 250th epochs, 40 original training samples and 40 generated samples for each class are randomly selected to visualize the 2-D embeddings. Figs. 7 and 8 provide the data distributions of the CNN-ResNet18 and ANN-ResNet18 models, respectively. Figs. 7(a) and 8(a) show that the features extracted from the generated samples are randomly distributed at the beginning of training. Thus, these generated samples cannot be used to train A^2NN models. After a warm-up process, the data distributions of most classes of generated samples are similar to those of the original training samples. At this point, the generated samples start to have practical significance and can be used to augment the training database. As the generator is trained, the quality of the generated samples increases, as shown in Fig. 7(b). Notably, due to the introduction of L_D, the generated sample distribution obtained by the ANN-ResNet18 model is also similar to the original sample distribution, as shown in Fig. 8(b).

To evaluate the convergence of the proposed knowledge matching-based generative learning method, the training loss of the generator and the accuracy of the knowledge sources are visualized in TensorBoardX by using the generated samples of the UCM dataset as training samples. The details of this process are shown in Fig. 9.

As shown in Fig. 9(a), the loss value of L_G decreases quickly during the training phase. With the introduction of L_D, the loss $L_{\mathrm{CE_{GA}}}$ can be reduced further until convergence. Since the generated samples are primarily optimized by the classification boundary and the distribution information of the well-trained CNN model, the accuracy of the CNN-ResNet18 model on the generated samples increases rapidly until convergence, as shown in Fig. 9(b). Similarly, the accuracy of the ANN-ResNet18 model on these generated samples increases quickly and achieves convergence because of the introduction of L_D.

In other words, the proposed knowledge matching-based generative learning method in our training strategy can generate effective samples for augmenting the training database without using the real samples or adversarial processes, thus alleviating the reduced accuracy of the A^2NN models.

G. Ablation Studies

To evaluate the effectiveness of the different components of the proposed GHKD training strategy, a series of ablation experiments are conducted with different training strategies

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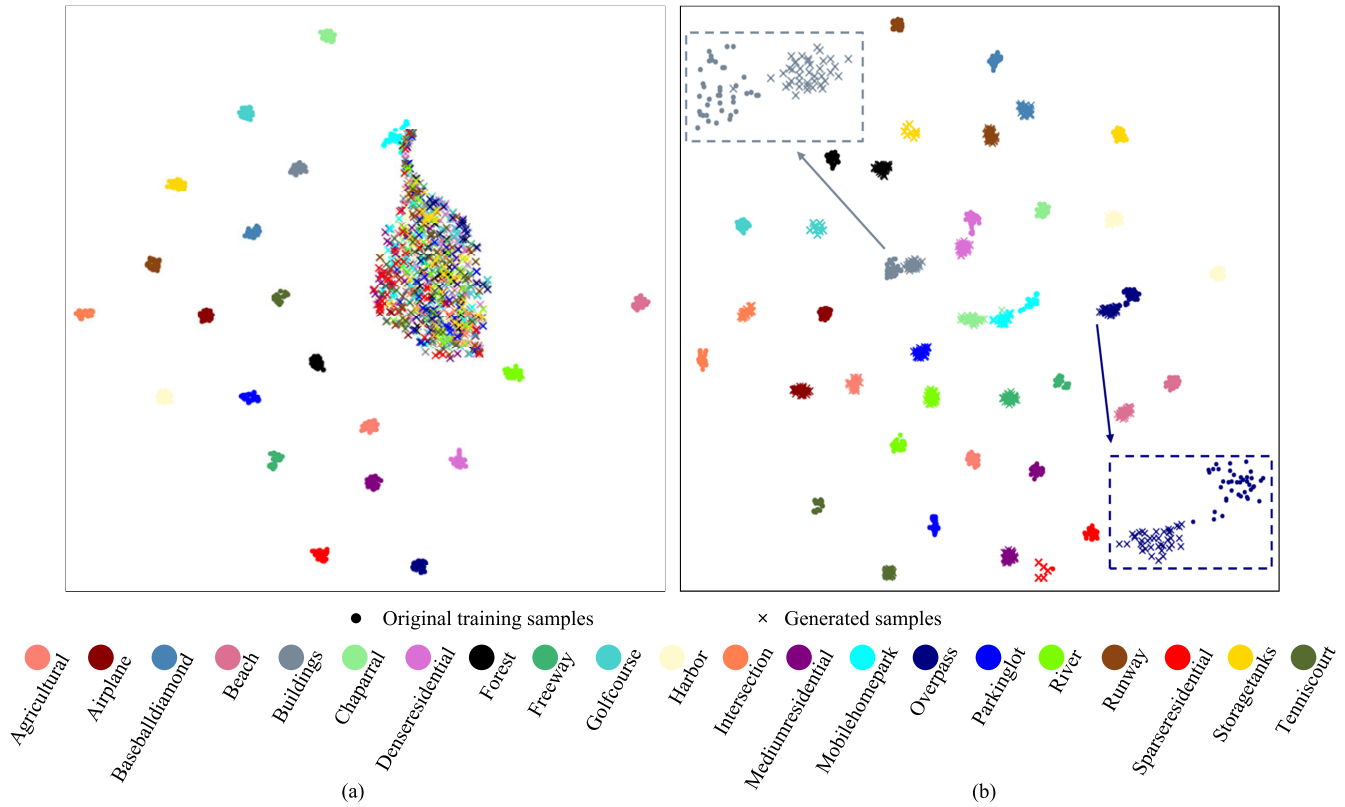

Fig. 8. Two-dimensional embedding of original training samples and generated samples on the UCM dataset with the ANN-ResNet18 model. (a) Epoch 1. (b) Epoch 250.

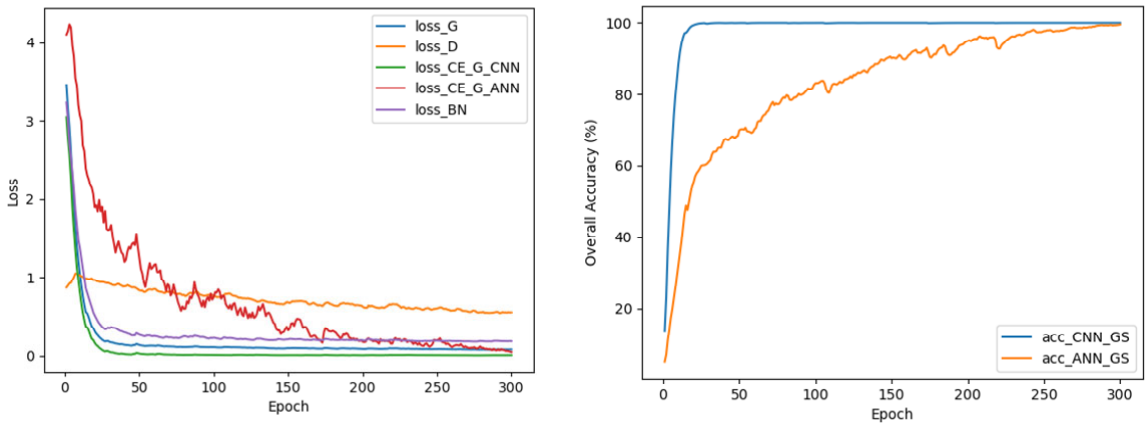

Fig. 9. Training visualization of the proposed knowledge matching-based generative learning method on the UCM dataset. (a) Training loss of the generator. (b) Accuracy of the knowledge sources by using the generated samples.

on the UCM and SIRI-WHU datasets. All experiments are repeated five times to verify the reliability of the classification results. The OAs of the A^{2}NN-ResNet18 model trained by different strategies are displayed in Table V.

Compared with the A^{2}NN model trained by the hard labels of the real-training samples S, the classification performance of the A^{2}NN model is noticeably enhanced when the soft labels of the teacher models are introduced into the training phase. This is because the well-trained teacher models can use their prior knowledge to guide the A^{2}NN model. Among the three variants, introducing the teacher CNN model for KD in the training phase brings the greatest performance improvement. This is because of the excellent feature extraction ability and classification performance of the well-trained CNN model, while the ANN model has a certain gap with the CNN model in classification performance. In particular, using the same real samples S as the training database for the teacher ANN model has an adverse effect on the knowledge transfer effect of the teacher CNN model.

Moreover, we design five variants for ablation studies when introducing generated samples as a new training database on this basis. First, when only one teacher model is introduced, using the mixed samples $\acute{S}$ obtained by mixing real samples S and generated samples $\hat{S}$ as the training database for

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TABLE V
ABLATION STUDY ON THE PROPOSED GHKD TRAINING STRATEGY [OA ± STD (%)]

Model	Training Strategy		Datasets	
			UCM	SIRI-WHU
A^2NN-ResNet18		Hard Labels	92.62±0.58	85.59±0.92
	+KD	Hard Labels + Soft Labels of Teacher CNN with S	95.19±0.39	89.38±0.34
		Hard Labels + Soft Labels of Teacher ANN with S	94.67±0.55	88.29±0.50
		Hard Labels + Soft Labels of Teacher CNN with S + Soft Labels of Teacher ANN with S	95.10±0.27	89.23±0.42
	+KD+$\hat{S}$	Hard Labels + Soft Labels of Teacher CNN with Ś	94.95±0.85	89.58±0.24
		Hard Labels + Soft Labels of Teacher ANN with Ś	94.62±0.13	87.21±0.48
		Hard Labels + Soft Labels of Teacher CNN with $\hat{S}$ + Soft Labels of Teacher ANN with $\hat{S}$	92.81±0.52	85.78±0.26
		Hard Labels + Soft Labels of Teacher CNN with Ś + Soft Labels of Teacher ANN with Ś	95.00±0.61	89.21±0.22
		Hard Labels + Soft Labels of Teacher CNN with $\hat{S}$ + Soft Labels of Teacher ANN with S	94.24±0.54	87.46±0.58
	GHKD (ours): Hard Labels + Soft Labels of Teacher CNN with S + Soft Labels of Teacher ANN with $\hat{S}$		95.62±0.27	89.61±0.24

knowledge transfer cannot ensure that the performance of the A^2NN model can be effectively improved. This is because expanding the training database with generated samples $\hat{S}$ for teacher CNN cannot solve the knowledge not being transmitted smoothly caused by the difference in weight distribution between CNN and A^2NN. Expanding the training database with generated samples $\hat{S}$ for teacher ANN cannot eliminate the gap in feature extraction ability between ANN and CNN. Second, using only the generated samples $\hat{S}$ as the training database for KD cannot effectively improve the performance of the A^2NN model. This is because the proposed knowledge matching-based generative learning method does not use real samples and the adversarial process during the training phase. Although the generated samples $\hat{S}$ and real samples S have similar distributions, the generated samples $\hat{S}$ cannot cover all the feature spaces of the real samples S. Therefore, the generated samples $\hat{S}$ can only be used to augment the training database and cannot be used to replace the original training samples completely. In addition, as shown in Table V, when the real samples S are used as the training database, the performance of the A^2NN model obtained by distillation using the CNN model as the teacher model is higher than that of the A^2NN model obtained by distillation using the ANN model as the teacher model. When the generated samples $\hat{S}$ are used as the training database, the ANN model is more suitable to be used as the teacher model than the CNN model. When both teacher CNN and teacher ANN use the mixed samples Ś as the training database, the performance of knowledge transfer is affected due to the gap in the classification performance of the two teachers. These results corroborate the rationality of the proposed hybrid KD method.

Finally, the A^2NN model trained by the proposed GHKD strategy achieves the best classification results among the above training strategies. These results show that the proposed GHKD training strategy is an effective approach for improving the performance of the A^2NN model.

VI. CONCLUSION

This article proposes a basic network that is suitable for deployment on edge devices, the A^2NN, and a GHKD training strategy to obtain high-performance A^2NN models for onboard RSSC. The A^2NN has a low resource overhead because it replaces all standard convolution kernels with adder kernels. The GHKD training strategy combines a hybrid KD method and a knowledge-matching-based generative learning method. The hybrid KD method utilizes the weight distribution knowledge in a well-trained ANN model to guide the high-performance classification knowledge in the well-trained CNN model to an A^2NN model that consists entirely of adder kernels. The knowledge-matching-based generative learning method generates effective samples by matching the knowledge about the feature distributions of the real samples in the knowledge sources to augment the training database for hybrid KD. The results of experiments on six public RSSC datasets indicate that the proposed GHKD training strategy can significantly improve the performance of A^2NN models for RSSC tasks and that the proposed A^2NN models achieve comparable or even better performance than the CNN models on most datasets. The well-trained A^2NN model obtained by the proposed training strategy, which has excellent performance and resource efficiency, can be directly deployed during the inference phase for onboard RSSC. In future work, we will attempt to further improve the generalization ability of the A^2NN models when faced with noise interference and distribution shifts, and explore a low-bit quantization algorithm for the A^2NN model to meet the needs of on-board real-time RSSC.

REFERENCES

[1] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[2] D. Hong, N. Yokoya, G.-S. Xia, J. Chanussot, and X. X. Zhu, "X-ModalNet: A semi-supervised deep cross-modal network for classification of remote sensing data," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 12–23, Sep. 2020.

[3] X. Du, X. Zheng, X. Lu, and A. A. Doudkin, "Multisource remote sensing data classification with graph fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10062–10072, Dec. 2021.

[4] W. Boulila, M. Sellami, M. Driss, M. Al-Sarem, M. Safaei, and F. A. Ghaleb, "RS-DCNN: A novel distributed convolutional-neural-networks based-approach for big remote-sensing image classification," *Comput. Electron. Agricult.*, vol. 182, Mar. 2021, Art. no. 106014.

[5] H. Zhu et al., "A spatial-channel progressive fusion ResNet for remote sensing classification," *Inf. Fusion*, vol. 70, pp. 72–87, Jun. 2021.

[6] W. Zhou, S. Newsam, C. Li, and Z. Shao, "PatternNet: A benchmark dataset for performance evaluation of remote sensing image retrieval," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 197–209, Nov. 2018.

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[7] X. Cao, J. Yao, Z. Xu, and D. Meng, "Hyperspectral image classification with convolutional neural network and active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4604–4616, Jul. 2020.

[8] N. Zhang, X. Wei, H. Chen, and W. Liu, "FPGA implementation for CNN-based optical remote sensing object detection," *Electronics*, vol. 10, no. 3, p. 282, Jan. 2021.

[9] H. Mahendra, S. Mallikarjunaswamy, G. Siddesh, M. Komala, and N. Sharmila, "Evolution of real-time onboard processing and classification of remotely sensed data," *Indian J. Sci. Technol.*, vol. 13, no. 20, pp. 2010–2020, May 2020.

[10] C. Vineyard, W. Severa, M. Kagie, A. Scholand, and P. Hays, "A resurgence in neuromorphic architectures enabling remote sensing computation," in *Proc. IEEE Space Comput. Conf. (SCC)*, Jul. 2019, pp. 33–40.

[11] S. Liu and W. Luk, "Towards an efficient accelerator for DNN-based remote sensing image segmentation on FPGAs," in *Proc. 29th Int. Conf. Field Program. Log. Appl. (FPL)*, Sep. 2019, pp. 187–193.

[12] C. Gonzalez, S. Bernabe, D. Mozos, and A. Plaza, "FPGA implementation of an algorithm for automatically detecting targets in remotely sensed hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 9, pp. 4334–4343, Sep. 2016.

[13] A. J. Plaza, "Special issue on architectures and techniques for real-time processing of remotely sensed images," *J. Real-Time Image Process.*, vol. 4, no. 3, pp. 191–193, Aug. 2009.

[14] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[15] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[16] B. Jacob et al., "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2704–2713.

[17] H. Li, A. Kadav, I. Durdanovic, H. Samet, and H. P. Graf, "Pruning filters for efficient ConvNets," 2016, *arXiv:1608.08710*.

[18] S. Han, H. Mao, and W. J. Dally, "Deep compression: Compressing deep neural networks with pruning, trained quantization and Huffman coding," 2015, *arXiv:1510.00149*.

[19] S. Gupta, A. Agrawal, K. Gopalakrishnan, and P. Narayanan, "Deep learning with limited numerical precision," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 1737–1746.

[20] M. Horowitz, "1.1 Computing's energy problem (and what we can do about it)," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2014, pp. 10–14.

[21] Y. Wang et al., "AdderNet and its minimalist hardware design for energy-efficient artificial intelligence," 2021, *arXiv:2101.10015*.

[22] M. V. Valueva, N. N. Nagornov, P. A. Lyakhov, G. V. Valuev, and N. I. Chervyakov, "Application of the residue number system to reduce hardware costs of the convolutional neural network implementation," *Math. Comput. Simul.*, vol. 177, pp. 232–243, Nov. 2020.

[23] H. Chen et al., "AdderNet: Do we really need multiplications in deep learning?" in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1468–1477.

[24] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.

[25] C. Shen, M. Xue, X. Wang, J. Song, L. Sun, and M. Song, "Customizing student networks from heterogeneous teachers via adaptive knowledge amalgamation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3504–3513.

[26] Z. Zheng, X. Yang, Z. Yu, L. Zheng, Y. Yang, and J. Kautz, "Joint discriminative and generative learning for person re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2138–2147.

[27] H. Li, H. Gu, Y. Han, and J. Yang, "Object-oriented classification of high-resolution remote sensing imagery based on an improved colour structure code and a support vector machine," *Int. J. Remote Sens.*, vol. 31, no. 6, pp. 1453–1470, Mar. 2010.

[28] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[29] Q. Zhu, Y. Zhong, B. Zhao, G.-S. Xia, and L. Zhang, "Bag-of-visual-words scene classifier with local and global features for high spatial resolution remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 6, pp. 747–751, Jun. 2016.

[30] B. Zhao, Y. Zhong, G.-S. Xia, and L. Zhang, "Dirichlet-derived multiple topic scene classification model for high spatial resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2108–2123, Apr. 2016.

[31] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.

[32] Q. Wang, W. Huang, Z. Xiong, and X. Li, "Looking closer at the scene: Multiscale representation learning for remote sensing image scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 4, pp. 1414–1428, Apr. 2022.

[33] F. Li, R. Feng, W. Han, and L. Wang, "High-resolution remote sensing image scene classification via key filter bank based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8077–8092, Nov. 2020.

[34] S. Wang, Y. Guan, and L. Shao, "Multi-granularity canonical appearance pooling for remote sensing scene classification," *IEEE Trans. Image Process.*, vol. 29, pp. 5396–5407, 2020.

[35] Y. Bazi, L. Bashmal, M. M. A. Rahhal, R. A. Dayil, and N. A. Ajlan, "Vision transformers for remote sensing image classification," *Remote Sens.*, vol. 13, no. 3, p. 516, Feb. 2021.

[36] P. Lv, W. Wu, Y. Zhong, F. Du, and L. Zhang, "SCViT: A spatial-channel feature preserving vision transformer for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4409512.

[37] J. Wang, H. Chen, L. Ma, L. Chen, X. Gong, and W. Liu, "Sphere loss: Learning discriminative features for scene classification in a hyperspherical feature space," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, 56018195601819.

[38] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[39] H. Huang et al., "IntroVAE: Introspective variational autoencoders for photographic image synthesis," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 1–12.

[40] M. Qi, J. Qin, Y. Wu, and Y. Yang, "Imitative non-autoregressive modeling for trajectory forecasting and imputation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12736–12745.

[41] N. Yu, K. Li, P. Zhou, J. Malik, L. Davis, and M. Fritz, "Inclusive GAN: Improving data and minority coverage in generative models," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 377–393, doi: 10.1016/j.neucom.2019.06.043.

[42] A. Ali-Gombe and E. Elyan, "MFC-GAN: Class-imbalanced dataset classification using multiple fake class generative adversarial network," *Neurocomputing*, vol. 361, pp. 212–221, Oct. 2019.

[43] G. Fang, J. Song, C. Shen, X. Wang, D. Chen, and M. Song, "Data-free adversarial distillation," 2019, *arXiv:1912.11006*.

[44] D. Lin, K. Fu, Y. Wang, G. Xu, and X. Sun, "MARTA GANs: Unsupervised representation learning for remote sensing image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2092–2096, Nov. 2017.

[45] Y. Yu, X. Li, and F. Liu, "Attention GANs: Unsupervised deep feature learning for aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 519–531, Jan. 2020.

[46] A. Ma, N. Yu, Z. Zheng, Y. Zhong, and L. Zhang, "A supervised progressive growing generative adversarial network for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618818.

[47] F. H. K. dos Santos Tanaka and C. Aranha, "Data augmentation using GANs," 2019, *arXiv:1904.09135*.

[48] H. Chen et al., "Data-free learning of student networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3514–3522.

[49] S. Xu et al., "Generative low-bitwidth data free quantization," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 1–17, doi: 10.1007/978-3-030-58610-2_1.

[50] J. Gou, B. Yu, S. J. Maybank, and D. Tao, "Knowledge distillation: A survey," *Int. J. Comput. Vis.*, vol. 129, no. 6, pp. 1789–1819, Jun. 2021.

[51] S. Zagoruyko and N. Komodakis, "Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer," 2016, *arXiv:1612.03928*.

[52] J. Yim, D. Joo, J. Bae, and J. Kim, "A gift from knowledge distillation: Fast optimization, network minimization and transfer learning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4133–4141.

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[53] J. Song et al., "Tree-like decision distillation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13488–13497.
[54] Y. Liu et al., "Knowledge distillation via instance relationship graph," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7096–7104.
[55] W. Park, D. Kim, Y. Lu, and M. Cho, "Relational knowledge distillation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3967–3976.
[56] B. Heo, M. Lee, S. Yun, and J. Y. Choi, "Knowledge transfer via distillation of activation boundaries formed by hidden neurons," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, vol. 33, no. 1, 2019, pp. 3779–3787.
[57] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.
[58] A. Paszke et al., "Automatic differentiation in PyTorch," in *Proc. NIPS Workshop*, 2017, pp. 1–4.
[59] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, Nov. 2010, pp. 270–279.
[60] G. Sheng, W. Yang, T. Xu, and H. Sun, "High-resolution satellite scene classification using a sparse coding based multiple feature combination," *Int. J. Remote Sens.*, vol. 33, no. 8, pp. 2395–2412, Apr. 2012.
[61] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2321–2325, Nov. 2015.
[62] L. Zhao, P. Tang, and L. Huo, "Feature significance-based multibag-of-visual-words model for remote sensing image scene classification," *J. Appl. Remote Sens.*, vol. 10, no. 3, Jul. 2016, Art. no. 035004.
[63] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.
[64] K. Xu, P. Deng, and H. Huang, "Vision transformer: An excellent teacher for guiding small networks in remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618715.
[65] K. Xu, H. Huang, and P. Deng, "Remote sensing image scene classification based on global–local dual-branch structure model," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[66] B. Li, Y. Guo, J. Yang, L. Wang, Y. Wang, and W. An, "Gated recurrent multiattention network for VHR remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5606113, doi: 10.1109/TGRS.2021.3093914.
[67] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[68] P. Liznerski, L. Ruff, R. A. Vandermeulen, B. J. Franks, M. Kloft, and K.-R. Müller, "Explainable deep one-class classification," 2020, *arXiv:2007.01760*.
[69] Y. Hu, X. Huang, X. Luo, J. Han, X. Cao, and J. Zhang, "Variational self-distillation for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5627313.
[70] M. Courbariaux, Y. Bengio, and J.-P. David, "BinaryConnect: Training deep neural networks with binary weights during propagations," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 1–9.
[71] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," 2016, *arXiv:1608.03983*.
[72] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[73] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 1–27, 2008.

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